

ASML Holding N.V.

Technology and Supply Chain Report — Fiscal Year 2023

Year Ending December 31, 2023

1. Company Overview and Strategic Position

ASML Holding N.V. is the world's sole supplier of extreme ultraviolet (EUV) lithography systems and holds a dominant position in deep ultraviolet (DUV) lithography. Headquartered in Veldhoven, Netherlands, ASML develops, produces, and services the photolithography machines used by semiconductor manufacturers to print integrated circuit patterns onto silicon wafers.

EUV lithography is a prerequisite technology for manufacturing chips at process nodes below 7 nanometers. Because no other company in the world produces EUV systems, ASML occupies a unique and irreplaceable position in the global semiconductor supply chain. ASML's EUV systems are used by TSMC, Samsung Foundry, and Intel Foundry Services — the three leading advanced semiconductor manufacturers globally.

2. Key Customers

TSMC is ASML's largest customer globally, purchasing EUV and DUV systems for its fabrication facilities in Taiwan, the United States (Arizona), and Japan (Kumamoto). TSMC's aggressive capacity expansion programme, driven by demand from Apple, Nvidia, AMD, and Qualcomm, requires continuous procurement of ASML's most advanced EUV systems. TSMC has placed long-term equipment orders with ASML that extend several years into the future to secure capacity in ASML's production queue.

Samsung Foundry is ASML's second-largest customer. Samsung uses EUV systems to manufacture chips for Qualcomm, Google's TPU programme, and other fabless customers. Samsung is also Apple's supplier for OLED display panels and DRAM memory, though Samsung does not manufacture Apple's primary SoC chips — that role belongs exclusively to TSMC.

Customer	Primary System Type	Primary Use	Estimated Share of ASML Revenue
TSMC	EUV (High-NA & standard)	Apple, Nvidia, AMD chips	~35%
Samsung Foundry	EUV & DUV	Qualcomm, Google TPU	~25%
Intel Foundry	EUV (High-NA)	Intel 18A process node	~15%
Other chipmakers	DUV primarily	Mature node production	~25%

3. EUV Technology and the Supply Chain Chokepoint

An EUV lithography system contains over 100,000 components sourced from more than 800 suppliers across 16 countries. Key subsystem suppliers include Carl Zeiss SMT (Germany) for the optical lenses and mirrors — which are so precise they must be manufactured to atomic-level tolerances — Cymer (a subsidiary of ASML itself, formerly US-based) for the light source, and VDL Groep (Netherlands) for the mechanical infrastructure.

The complexity and precision required to manufacture an EUV system means that ASML's production capacity is inherently limited. In fiscal year 2023, ASML shipped 53 EUV systems, representing approximately USD 8.4 billion in EUV revenue. ASML is developing a next-generation High-NA EUV system (the TWINSCAN EXE:5000) with a higher numerical aperture that enables even smaller feature sizes. TSMC and Intel have placed orders for High-NA EUV systems for their next-generation process nodes.

4. Export Controls and Geopolitical Risk

ASML operates under strict export control regimes administered by the Dutch, US, and Japanese governments. Since 2019, ASML has been prohibited from exporting EUV systems to China under pressure from the United States government. In October 2023, the Dutch government expanded export restrictions to include certain DUV systems (specifically the TWINSCAN NXT:2000i and more advanced models), significantly limiting ASML's addressable market in China.

These export restrictions directly affect SMIC and other Chinese chipmakers, preventing them from accessing sub-7nm process technology. This benefits TSMC, Samsung, and Intel Foundry competitively, but represents a revenue risk for ASML. China accounted for approximately 26% of ASML's total revenue in fiscal year 2023, primarily through DUV system sales. Further export restriction expansion could reduce ASML's China revenue materially.

5. Financial Summary

Metric	FY2023	FY2022	Change
Net Sales	EUR 27.6B	EUR 21.2B	+30.2%
EUV System Revenue	EUR 8.4B	EUR 6.4B	+31.3%
DUV System Revenue	EUR 12.1B	EUR 9.2B	+31.5%
Gross Margin	51.3%	50.5%	+0.8 pp
Net Income	EUR 7.8B	EUR 5.6B	+39.3%
EUV Systems Shipped	53 units	40 units	+32.5%

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